

ANOVA-REGRESSION

Part C

One way ANOVA

- Response variable: Brinell hardness
- Factor: Quenching procedure
- Levels: T1 T2 T3
- Replicates: 3
- Order of runs: random
- Observations:
 - T1: 70, 70, 90
 - T2: 120, 170, 120
 - T3: 140, 180, 180

Model

- Let us assume the following model

$$y_{ij} = \mu_i + \varepsilon_{ij} \quad \text{with} \quad \varepsilon_{ij} \sim NID(0, \sigma^2)$$

$$i = 1, 3 \quad j = 1, 3$$

- Or, equivalently

$$y_{ij} = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon_{ij}$$

$$x_1 = \{0, 1\} \quad x_2 = \{0, 1\} \quad x_3 = \{0, 1\}$$

In matrix form

$$\begin{Bmatrix} 70 \\ 70 \\ 90 \\ 120 \\ 170 \\ 120 \\ 140 \\ 180 \\ 180 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{Bmatrix} + \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{22} \\ \varepsilon_{23} \\ \varepsilon_{31} \\ \varepsilon_{32} \\ \varepsilon_{33} \end{Bmatrix}$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Estimate of Model Parameters

- Let us write the matrix $\mathbf{X}$ in the following way

$$\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2 \ \mathbf{X}_3] \rightarrow \mathbf{X}^T = \begin{bmatrix} \mathbf{X}_1^T \\ \mathbf{X}_2^T \\ \mathbf{X}_3^T \end{bmatrix} \quad \text{Note that } \forall i \neq j \ \mathbf{X}_i^T \mathbf{X}_j = 0$$

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} \mathbf{X}_1^T \mathbf{X}_1 & 0 & 0 \\ 0 & \mathbf{X}_2^T \mathbf{X}_2 & 0 \\ 0 & 0 & \mathbf{X}_3^T \mathbf{X}_3 \end{bmatrix} \rightarrow (\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} \frac{1}{\mathbf{X}_1^T \mathbf{X}_1} & 0 & 0 \\ 0 & \frac{1}{\mathbf{X}_2^T \mathbf{X}_2} & 0 \\ 0 & 0 & \frac{1}{\mathbf{X}_3^T \mathbf{X}_3} \end{bmatrix}$$

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{Bmatrix} \mathbf{X}_1^T \mathbf{y} / \mathbf{X}_1^T \mathbf{X}_1 \\ \mathbf{X}_2^T \mathbf{y} / \mathbf{X}_2^T \mathbf{X}_2 \\ \mathbf{X}_3^T \mathbf{y} / \mathbf{X}_3^T \mathbf{X}_3 \end{Bmatrix} = \begin{Bmatrix} y_{1\cdot} / 3 \\ y_{2\cdot} / 3 \\ y_{3\cdot} / 3 \end{Bmatrix} = \begin{Bmatrix} \frac{70+70+90}{3} \\ \frac{120+170+120}{3} \\ \frac{140+180+180}{3} \end{Bmatrix} = \begin{Bmatrix} 76.\bar{6} \\ 136.\bar{6} \\ 166.\bar{6} \end{Bmatrix}$$

$$\begin{aligned} SS_{REG}^* &= \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y} = \hat{\beta}_1 \mathbf{X}_1^T \mathbf{y} + \hat{\beta}_2 \mathbf{X}_2^T \mathbf{y} + \hat{\beta}_3 \mathbf{X}_3^T \mathbf{y} = \\ &= \bar{y}_{1\cdot} y_{1\cdot} + \bar{y}_{2\cdot} y_{2\cdot} + \bar{y}_{3\cdot} y_{3\cdot} = \frac{\sum_{j=1,3} y_{j\cdot}^2}{n} = \\ &= 76.\bar{6} * 230 + 136.\bar{6} * 410 + 166.\bar{6} * 500 = 157000 \end{aligned}$$

$$SS_{TOT}^* = \mathbf{y}^T \mathbf{y} = \sum_{i,j} y_{ij}^2 = 160000$$

Source	SS	<i>df</i>	MS	F	p
Regression*	157000.0	3	5233.3	104.6	0
Error	3000.0	6	500.0		
Total*	160000.0	9			

Quite significant model (we should check the hypotheses). However the significance could have been given by the **constant term** of the model alone

- **Reduced Model (without the factor terms)**

$$y_{ij} = \mu + \varepsilon_{ij}$$

$$\mathbf{X} = \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} \equiv \mathbf{1} \quad \mathbf{1}^T = [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1] \quad (\mathbf{1}^T \mathbf{1}) = 9 \quad (\mathbf{1}^T \mathbf{1})^{-1} = 1/9$$

$$\hat{\beta}_0 = \hat{\mu} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = (\mathbf{1}^T \mathbf{1})^{-1} \mathbf{1}^T \mathbf{y} = \frac{\sum_{ij} y_{ij}}{9} = \bar{y}_{..}$$

$$SS_{REG}^* = \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y} = \bar{y}_{..} \sum_{ij} y_{ij} = \frac{y_{..}^2}{9} = \frac{(230 + 410 + 500)^2}{9} = 144400$$

Sources	SS	<i>df</i>	MS	F	p
Constant	144400.0	1	144400.0	288.8	0
Regression Corrected	12600.0	2	6300.0	12.6	0
Error	3000.0	6	500.0		
Total*	160000.0	9			

p-value for the «corrected regression» term is 0.0071

The «corrected regression» term is significant

Usually we write the ANOVA table without the constant term

Source	SS	<i>df</i>	MS	F	p
Regression (corrected: without the constant)	12600.0	2	6300.0	12.6	0
Error	3000.0	6	500.0		
Total (corrected)	15600.0	8			

We conclude that the regression (corrected) is highly significant

Hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$

$$\mathbf{H} = \begin{bmatrix} 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \end{bmatrix}$$

remember? $\rightarrow \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $Tr(\mathbf{H}) = p = 3$

IF we had used the model with the constant?

X matrix:

1	1	0
1	1	0
1	1	0
1	0	1
1	0	1
1	0	1
1	-1	-1
1	-1	-1
1	-1	-1

$$y = \mu + \tau_i + \varepsilon \quad \text{with} \quad \tau_3 = -\tau_1 - \tau_2$$

$$y = \mu + \tau_1 x_1 + \tau_2 x_2 + \varepsilon$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$$

$$(x_1 = 1; x_2 = 0) \quad \hat{y}_1 = 126.67 - 50 = 76.67 \quad \hat{\beta}_0 = 126.67$$

$$(x_1 = 0; x_2 = 1) \quad \hat{y}_2 = 126.67 + 10 = 136.67 \quad \hat{\beta}_1 = -50$$

$$(x_1 = -1; x_2 = -1) \quad \hat{y}_3 = 126.67 + 50 - 10 = 166.67 \quad \hat{\beta}_2 = 10$$



Note the coding system

- Hat Matrix

The same matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Two factors ANOVA

- Response variable: Brinell Hardness
- Factor 1: Quenching procedure
- Factor 2: Operator
- Levels for 1: T1 T2 T3
- Levels for 2: O1 O2
- Number of replicates: 2, Number of treatments=6
- Design: Full Factorial
- Run order: Random
- Observations N= 12
 - T1-O1: 70; 80 T1-O2: 75; 77
 - T2-O1: 120; 130 T2-O2: 135; 153
 - T3-O1: 140; 159 T3-O2: 165; 170

Fit Model (by JMP)

Fit Model builds a **X** matrix of full rank using the usual constraints on the model terms

$$y = \mu + \tau_i + \beta_j + \tau\beta_{ij} + \varepsilon$$

$$y = \mu x_0 + \tau_1 x_1 + \tau_2 x_2 + \beta_1 x_3 + \tau\beta_{11} x_4 + \tau\beta_{21} x_5 + \varepsilon$$

μ τ_1 τ_2 β_1 $\tau\beta_{11}$ $\tau\beta_{21}$

T1	O1	1	1	0	1	1	0
T1	O1	1	1	0	1	1	0
T1	O2	1	1	0	-1	-1	0
T1	O2	1	1	0	-1	-1	0
T2	O1	1	0	1	1	0	1
T2	O1	1	0	1	1	0	1
T2	O2	1	0	1	-1	0	-1
T2	O2	1	0	1	-1	0	-1
T3	O1	1	-1	-1	1	-1	-1
T3	O1	1	-1	-1	1	-1	-1
T3	O2	1	-1	-1	-1	1	1
T3	O2	1	-1	-1	-1	1	1

$$\tau_1 + \tau_2 + \tau_3 = 0$$

$$\beta_1 + \beta_2 = 0$$

$$\tau\beta_{11} + \tau\beta_{12} = 0 \rightarrow \tau\beta_{12} = -\tau\beta_{11}$$

$$\tau\beta_{21} + \tau\beta_{22} = 0 \rightarrow \tau\beta_{22} = -\tau\beta_{21}$$

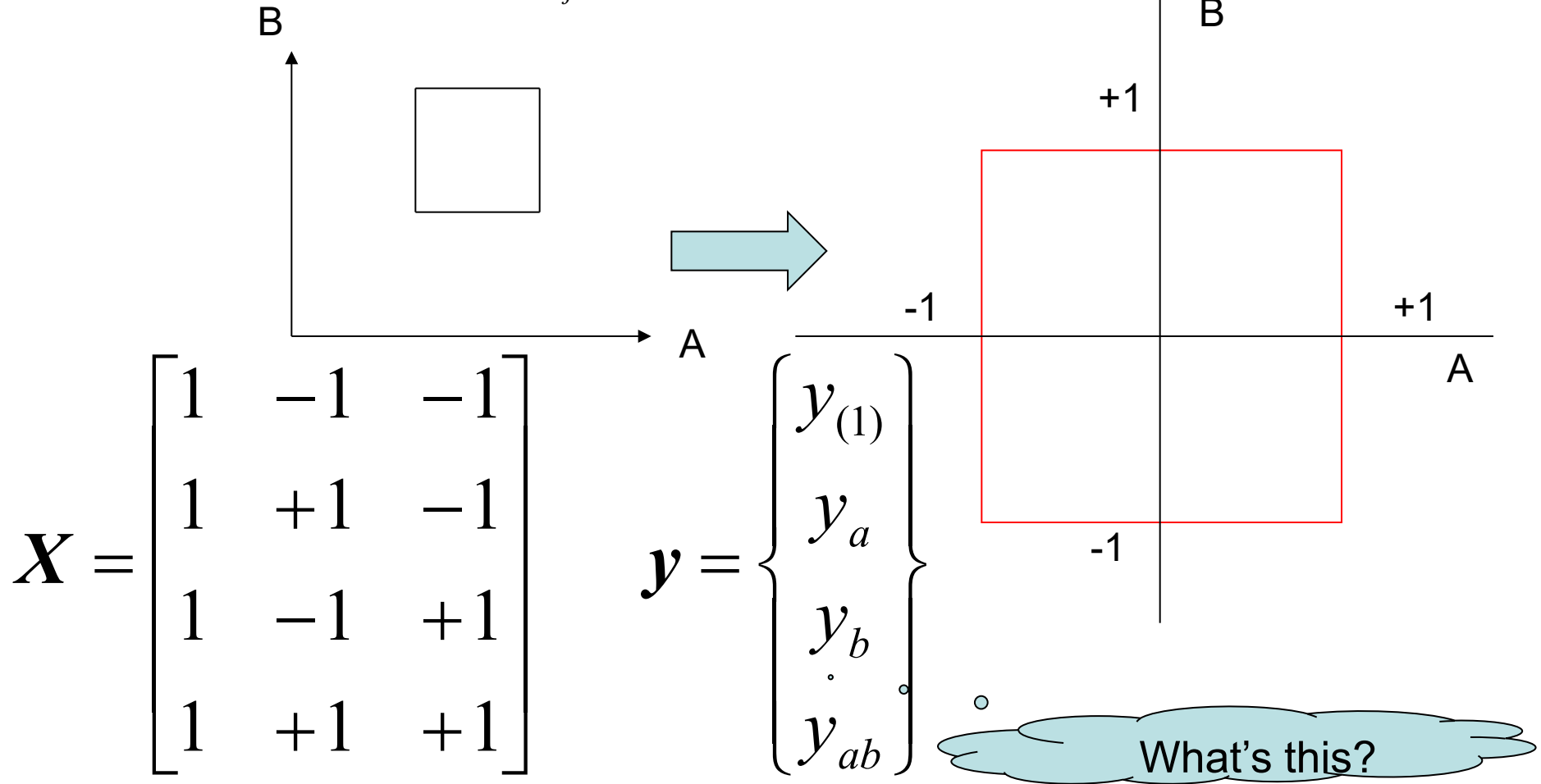
$$\tau\beta_{31} + \tau\beta_{32} = 0 \rightarrow \tau\beta_{32} = -\tau\beta_{31} = \tau\beta_{11} + \tau\beta_{21}$$

$$\tau\beta_{11} + \tau\beta_{21} + \tau\beta_{31} = 0 \rightarrow \tau\beta_{31} = -\tau\beta_{11} - \tau\beta_{21}$$

$$\tau\beta_{12} + \tau\beta_{22} + \tau\beta_{32} = 0 \rightarrow \tau\beta_{32} = \tau\beta_{11} + \tau\beta_{21}$$

2^k design, without replicates

$$y_{ij} = \beta_0 + \beta_A x_A + \beta_B x_B + \varepsilon$$



$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = N\mathbf{I} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

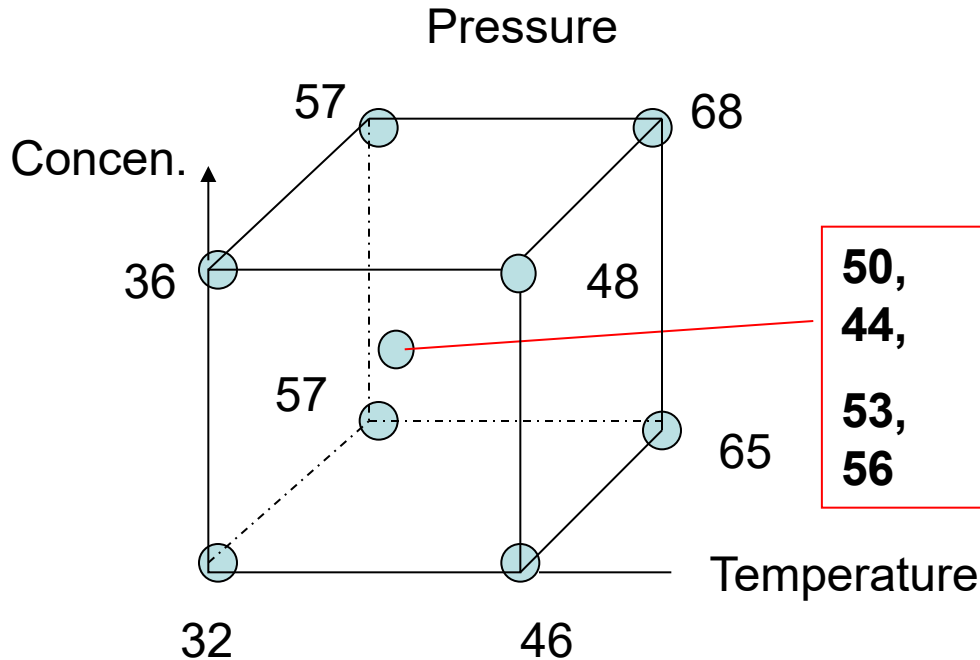
$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \rightarrow \text{Cov}(\hat{\beta}_i, \hat{\beta}_j) = 0$$

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \begin{Bmatrix} y_{(1)} \\ y_a \\ y_b \\ y_{ab} \end{Bmatrix} = \begin{Bmatrix} \frac{y_{(1)} + y_a + y_b + y_{ab}}{4} \\ \frac{-y_{(1)} + y_a - y_b + y_{ab}}{4} \\ \frac{-y_{(1)} - y_a + y_b + y_{ab}}{4} \end{Bmatrix}$$

$$EFF_{[\]} = \text{Cont}_{[\]} / n2^{k-1} = 2\hat{\beta}_{[\]}$$

Example 10.2

$$y_{ij} = \beta_0 + \beta_A x_A + \beta_B x_B + \beta_C x_C + \varepsilon$$



Response variable:
Reaction yield

A: Temp. (120, 160)

B: Pressure (40, 80)

C: Concentr. (15, 30)

Code the regressors

$$x_A = \frac{\text{Temperature} - 140}{20}$$

$$x_1 \in \{-1 \quad 1\}$$

$$x_B = \frac{\text{Pressure} - 60}{20}$$

$$x_2 \in \{-1 \quad 1\}$$

$$x_C = \frac{\text{Concentration} - 22.5}{7.5}$$

$$x_3 \in \{-1 \quad 1\}$$

Factorial

$$\mathbf{X} = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Central point

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 12 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 32 \\ 46 \\ 57 \\ 65 \\ 36 \\ 48 \\ 57 \\ 68 \\ 50 \\ 44 \\ 53 \\ 56 \end{bmatrix}$$

$$\mathbf{X}^T \mathbf{y} = \begin{bmatrix} 612 \\ 45 \\ 85 \\ 9 \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 1/12 & 0 & 0 & 0 \\ 0 & 1/8 & 0 & 0 \\ 0 & 0 & 1/8 & 0 \\ 0 & 0 & 0 & 1/8 \end{bmatrix} \begin{Bmatrix} 612 \\ 45 \\ 85 \\ 9 \end{Bmatrix} = \begin{Bmatrix} 51.0 \\ 5.625 \\ 10.625 \\ 1.125 \end{Bmatrix}$$

The experiment is still **orthogonal** (even if we added some central points, the columns of $\mathbf{X}$ are still orthogonals)

The central points does **NOT affect** the parameter estimates, except the constant parameter

The fitted regression model (Additive Model) is:

$$y = 51.0 + 5.625x_A + 10.625x_B + 1.125x_C$$

Try to use the full model: is it still orthogonal?

Example 10.3

- Let us suppose that one data is missing. The specimen of the **(1,1,1)** condition was damaged

$$X = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ \equiv & \equiv & \equiv & \equiv \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad y = \begin{Bmatrix} 32 \\ 46 \\ 57 \\ 65 \\ 36 \\ 48 \\ 57 \\ \equiv \\ 50 \\ 44 \\ 53 \\ 56 \end{Bmatrix}$$

The experiment **is not orthogonal**
($X^T X$) Matrix is not diagonal

- $\mathbf{X}^T\mathbf{X}$ Matrix

$$\begin{pmatrix} 11 & -1 & -1 & -1 \\ -1 & 7 & -1 & -1 \\ -1 & -1 & 7 & -1 \\ -1 & -1 & -1 & 7 \end{pmatrix}$$

- $(\mathbf{X}^T\mathbf{X})^{-1}$ Matrix

$$\begin{pmatrix} \frac{5}{52} & \frac{1}{52} & \frac{1}{52} & \frac{1}{52} \\ \frac{1}{52} & \frac{2}{13} & \frac{3}{104} & \frac{3}{104} \\ \frac{3}{104} & \frac{3}{104} & \frac{2}{13} & \frac{3}{104} \\ \frac{3}{104} & \frac{3}{104} & \frac{3}{104} & \frac{2}{13} \end{pmatrix}$$

The matrices are not orthogonal

The fitted regression model is:

$$\hat{y} = 51.25 + 5.75x_A + 10.75x_B + 1.25x_C \quad (\text{new model})$$

$$\hat{y} = 51.0 + 5.625x_A + 10.625x_B + 1.125x_C \quad (\text{old model})$$

$$\hat{y}_{-(1,1,1)} = 51.25 + 5.75 + 10.75 + 1.25 = 69 \quad (\text{old prediction} = 68.375)$$

Summing up

- With the Standard ANOVA model we can not make predictions for conditions (treatments) not used during the experimentation.
- Regression models allow us to overcome this problem if the Factors are quantitatatives